

Spectral Probing of the GigaPath Backbone

Jacobian Singular Value Analysis as a Dynamical Systems Diagnostic

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1 Motivation: Layers as a Dynamical System

We regard the depth-wise application of transformer blocks in the GigaPath backbone as a discrete-time dynamical system. Let $\mathbf{h}_l \in \mathbb{R}^{N \times d}$ denote the token-sequence hidden state at layer l , where N is the number of tokens (including the CLS token) and $d = 1536$ is the embedding dimension. Each transformer block defines the recurrence

$$\mathbf{h}_{l+1} = F_l(\mathbf{h}_l; \theta_l) = \mathbf{h}_l + G_l(\mathbf{h}_l; \theta_l), \quad (1)$$

where G_l is the residual branch (self-attention $\rightarrow$ layer-scale $\rightarrow$ MLP $\rightarrow$ layer-scale) and θ_l collects the parameters of block l . The total depth is $L = 40$.

Attractor hypothesis. After self-supervised pre-training the system (1) has converged to a regime in which input representations are mapped to a low-dimensional attractor. The geometry of this attractor is characterised by the *Lyapunov spectrum*: the rates at which infinitesimally nearby trajectories (i.e. nearby inputs) diverge or converge as they propagate through the layers.

2 Which Jacobian?

Two natural Jacobians can be defined at every layer:

Symbol	Definition	Measures
$\mathbf{J}_l^{(h)} = \frac{\partial \mathbf{h}_{l+1}}{\partial \mathbf{h}_l}$	State-space Jacobian	Sensitivity of output to <i>hidden state</i> perturbations
$\mathbf{J}_l^{(\theta)} = \frac{\partial \mathbf{h}_{l+1}}{\partial \theta_l}$	Parameter Jacobian	Sensitivity of output to <i>weight</i> perturbations

For Lyapunov analysis, use the state-space Jacobian $\mathbf{J}_l^{(h)}$. Lyapunov exponents characterise how nearby *trajectories* (hidden states produced by nearby inputs) diverge or converge. This is entirely governed by $\mathbf{J}_l^{(h)}$. The parameter Jacobian $\mathbf{J}_l^{(\theta)}$ measures Fisher-information-type sensitivities, which is a different (though related) diagnostic.

The residual structure (1) implies

$$\mathbf{J}_l^{(h)} = \mathbf{I} + \frac{\partial G_l}{\partial \mathbf{h}_l}, \quad (2)$$

so the Jacobian is a perturbation of the identity.

3 Singular Values and the Lyapunov Spectrum

3.1 Notation

Let $\sigma_i^{(l)}$ denote the i -th singular value of $\mathbf{J}_l^{(h)}$, i.e.

$$\sigma_i^{(l)} = \sigma_i(\mathbf{J}_l^{(h)}) = \sqrt{\lambda_i(\mathbf{J}_l^{(h)\top} \mathbf{J}_l^{(h)})}, \quad (3)$$

where $\lambda_i(\cdot)$ denotes the i -th eigenvalue (in decreasing order). In other words, $\sigma_i^{(l)}$ is the i -th singular value of the layer- l Jacobian, measuring the local expansion ($\sigma > 1$) or contraction ($\sigma < 1$) rate in the i -th principal direction at that layer.

- $\sigma_i^{(l)} > 1$: the i -th direction is **expanded** by layer l — information is amplified.
- $\sigma_i^{(l)} < 1$: the i -th direction is **contracted** — information is discarded.
- $\sigma_i^{(l)} \approx 1$: the i -th direction is approximately **preserved** (dynamical isometry).

3.2 Finite-depth Lyapunov exponents

The (finite-depth) Lyapunov exponents of the system are defined by the singular values of the composed mapping $\mathbf{J}^{[0:L]} = \prod_{l=0}^{L-1} \mathbf{J}_l^{(h)}$. Concretely, by the multiplicative ergodic theorem (Oseledets, 1968), the Lyapunov exponents are

$$\Lambda_i = \lim_{L \rightarrow \infty} \frac{1}{2L} \log \lambda_i \left(\prod_{l=0}^{L-1} \mathbf{J}_l^{(h)\top} \mathbf{J}_l^{(h)} \right). \quad (4)$$

For a finite-depth network, we replace the limit with a finite sum and define the *empirical Lyapunov exponents* as

$$\hat{\Lambda}_i = \frac{1}{L} \sum_{l=0}^{L-1} \log \sigma_i^{(l)}. \quad (5)$$

This is an approximation; strictly it assumes the singular-value directions align across layers, but it provides a useful summary statistic.

Key observables.

Observable	Interpretation
$\sigma_{\max}^{(l)} \gg 1$	Layer l strongly amplifies certain directions
$\sigma_{\min}^{(l)} \ll 1$	Layer l strongly contracts certain directions
$\kappa^{(l)} = \sigma_{\max}^{(l)} / \sigma_{\min}^{(l)} \gg 1$	High condition number; anisotropic geometry
All $\sigma_i^{(l)} \approx 1$	Dynamical isometry (ideal for signal propagation)
$\hat{\Lambda}_i > 0$	Net expansion over the full depth — chaos-like regime
$\hat{\Lambda}_i < 0$	Net contraction — attractor-like regime
$\hat{\Lambda}_i \approx 0$	Critical regime (edge of chaos)

3.3 Why the residual structure matters

From (2), $\mathbf{J}_l^{(h)} = \mathbf{I} + \partial G_l / \partial \mathbf{h}_l$, so $\sigma_i^{(l)} = 1 + \delta_i^{(l)}$ where $\delta_i^{(l)}$ are the singular-value perturbations introduced by the residual branch. The interesting diagnostic is the *deviation spectrum* $\{\delta_i^{(l)}\}$, or equivalently $\{\log \sigma_i^{(l)}\}$, plotted as a function of layer index l .

3.4 Input dependence

Because transformers use attention, $\mathbf{J}_l^{(h)}$ depends on the input $\mathbf{h}_l$, which itself depends on the data point. The spectral probe therefore produces a *per-sample* local Lyapunov spectrum. In practice, we aggregate by averaging singular values (or their logs) over a mini-batch.

4 Feasibility of the Attractor Hypothesis

The hypothesis is well-supported by prior work:

1. **Deep Information Propagation** (Schoenholz et al., 2017): at initialisation, deep networks must operate at the edge of chaos (all Lyapunov exponents ≈ 0) for trainability.
2. **Dynamical isometry** (Saxe et al., 2014; Pennington et al., 2018): well-trained networks have Jacobian singular values clustered near 1.
3. **Residual networks** (Yang & Schoenholz, 2018): residual connections bias $\mathbf{J}$ toward the identity, naturally promoting dynamical isometry.
4. **For a frozen pre-trained ViT** such as GigaPath, the expectation is that most $\sigma_i^{(l)} \approx 1$, with a few extreme values encoding the dominant feature-transformation modes. The largest singular values correspond to the directions the network *amplifies* (semantically important features), and the smallest to the directions it *discards*.

5 Efficient Computation

5.1 Dimensionality

With $N = 197$ tokens (196 patches + CLS) and $d = 1536$, the full Jacobian $\mathbf{J}_l^{(h)} \in \mathbb{R}^{Nd \times Nd}$ has dimension $302,592 \times 302,592$. Materialising or decomposing this matrix is infeasible.

5.2 Two modes of operation

We implement two complementary modes:

Mode 1: CLS-token-only (materialisable). If we restrict to the CLS token (the first row of the token sequence), the Jacobian reduces to

$$\mathbf{J}_l^{\text{cls}} = \frac{\partial [\mathbf{h}_{l+1}]_{\text{cls}}}{\partial [\mathbf{h}_l]_{\text{cls}}} \in \mathbb{R}^{d \times d}, \quad (6)$$

where $d = 1536$. This 1536×1536 matrix can be materialised directly via `torch.autograd.functional.jacobian` and decomposed with standard dense SVD in ~ 1 second on GPU. This gives a complete spectrum of all 1536 singular values.

Mode 2: Full-sequence matrix-free (via JVP/VJP). For the full $Nd \times Nd$ Jacobian, we avoid materialisation entirely. We define a `scipy.sparse.linalg.LinearOperator` that implements the matrix-vector products

$$\text{matvec: } \mathbf{u} = \mathbf{J} \mathbf{v} \quad \text{via} \quad \text{torch.func.jvp} \quad (7)$$

$$\text{rmatvec: } \mathbf{v} = \mathbf{J}^T \mathbf{u} \quad \text{via} \quad \text{torch.func.vjp} \quad (8)$$

and then call `scipy.sparse.linalg.svds` to obtain the top- k and bottom- k singular values via implicitly-restarted Lanczos bidiagonalisation. Each call to (7) or (8) has the cost of one forward or backward pass through a *single* transformer block.

5.3 Overall cost

Mode	Cost per layer	Output
CLS-only	$\sim d^2$ forward ops + dense SVD	All d singular values
Matrix-free, top/bottom k	$\sim (2k \cdot n_{\text{iter}})$ block fwd/bwd passes	Top- k and bottom- k singular values

Probing 5 evenly-spaced blocks with $k = 10$ in matrix-free mode requires $\sim 5 \times 2 \times 10 \times 20 = 2000$ block evaluations (assuming $n_{\text{iter}} \approx 20$ Lanczos steps), versus 40 blocks for one full forward pass. This is $\sim 50\times$ overhead *per probed sample*, so the probe should be run on a small batch (e.g. 1–4 samples) and not on every evaluation step. The CLS-only mode is far cheaper ($\sim 3\text{--}5\times$ one forward pass for 5 blocks) and is recommended for routine monitoring.

6 Implementation

6.1 SpectralProbe: standalone spectral estimator

Given a transformer block F_l and an activation tensor $\mathbf{h}_l$, the `SpectralProbe` estimates:

- **CLS mode:** materialises $\mathbf{J}_l^{\text{cls}} \in \mathbb{R}^{d \times d}$ via `torch.autograd.functional.jacobian`, then calls `torch.linalg.svdvals`.
- **Full mode:** wraps JVP/VJP in a `LinearOperator` and calls `scipy.sparse.linalg.svds` for top/bottom k .

6.2 SpectralBackboneEvaluator: integrated evaluator

`SpectralBackboneEvaluator` subclasses `SidebandBackboneEvaluator`. It reuses the existing hook infrastructure to capture intermediate activations at the probed blocks, then runs `SpectralProbe` on those activations after the forward pass. Results are stored in the `spectral_results` property, keyed by block index.

The evaluator is instantiated via `gigapath_backbone_evaluator` with name "GIGAPATH_SPECTRAL_BACKBONE_".

6.3 Pipeline entry point

In `pipelines.py`:

```
gigapath_backbone_evaluator(
    "GIGAPATH_BASELINE_BACKBONE_SPECTRAL_EVALUATOR",
    device='cuda',
)
```

This returns a `SpectralBackboneEvaluator` that probes 5 evenly-spaced blocks (indices 0, 10, 20, 30, 39) and estimates both the CLS-only full spectrum and the top/bottom 10 matrix-free singular values.

7 Interpretation Guide

Plot $\log \sigma_i^{(l)}$ versus layer index l for the top and bottom singular values. Expected patterns for a well-trained ViT:

1. **Most singular values near 1** (i.e. $\log \sigma \approx 0$): the network is approximately isometric, preserving the geometry of the representation.
2. **A few large singular values** ($\sigma \gg 1$): these correspond to directions the network selectively amplifies — likely the most semantically discriminative feature directions.
3. **A few small singular values** ($\sigma \ll 1$): these correspond to directions being aggressively compressed — the network is performing dimensionality reduction along these directions.
4. **Singular values converging as l increases**: this supports the attractor hypothesis — later layers are “closer” to the fixed point, with the Jacobian becoming more isometric.
5. **Condition number $\kappa^{(l)}$ decreasing with depth**: further evidence of convergence to an attractor with decreasing anisotropy.